When Will Quantum Computers Beat Classical Simulation of a Real Quantum Magnet? A Quantum-Classical Crossover for the Neutron Spectrum of α-$RuCl_3$

Anonymous Author(s)
Proceedings Track submission

***Abstract*. Claims that quantum computers will beat classical computers at simulating materials are usually made for toy models, at an accuracy the authors choose. We ask the question for a real material, with the accuracy set by an experiment. Our target is the neutron-scattering spectrum $S(\mathbf{q},\omega)$ of the Kitaev magnet α-$RuCl_3$, computed to the energy resolution of the instrument. We time exact classical simulations on clusters of up to 18 spins and matrix-product-state (MPS) simulations on cylinders. We measure the Trotter error of the matching quantum circuits against exact results and give the circuits to a surface-code resource estimator. For three published models of the material, a fault-tolerant quantum computer becomes faster than exact simulation at $N^* \approx 36$–38 spins. For the model of Winter *et al.* it then needs 0.25 M physical qubits at a physical error rate of 10^{-4}, or 1.7 M at 10^{-3}. At this point both machines need about a century, so the crossover marks where neither is practical, not where quantum simulation becomes useful. Preparing the ground state from the best simple starting state moves the crossover to $N^* \approx 92$–94. Even the layout with the fewest qubits needs 29–34 thousand physical qubits at an error rate of 10^{-4}. That is 29–34 times the 1000-qubit target of India's National Quantum Mission. The two-qubit error reported for the 25-qubit QpiAI-Indus system is 4 times above the surface-code threshold. We do not claim an advantage today. We state the assumptions behind every projection, and code and data will be made public.**

***Index Terms*. quantum advantage, resource estimation, Kitaev materials, neutron scattering, Trotter error**

Plain-language summary. Neutron beams show how the tiny magnets inside a crystal move. To compare these measurements with a model, we need a computer simulation, and it gets exponentially harder as the simulated piece of crystal grows. We estimate how large that piece must be before a future error-corrected quantum computer would finish faster than a classical computer, using the real accuracy of neutron experiments on the magnet α-$RuCl_3$. The answer is a few dozen magnetic atoms. Reaching it needs hundreds of thousands of physical qubits, far more than any machine built or planned so far.

I. Introduction

Simulating how quantum systems change in time is the use most often named for quantum computers. Yet most studies of when quantum computers win use a model system and an accuracy chosen by the authors. Yoshioka *et al.* [1] found the crossover point for ground-state energies of two-dimensional models. Sun *et al.* [2] did the same for the dynamics of the mixed-field Ising model, against tensor-network and variational Monte Carlo methods. Two things are still missing for a result a materials scientist can use: a real material whose model cannot be solved exactly, and an accuracy set by a real measurement rather than by a free parameter ε.

α-$RuCl_3$ is the leading candidate for Kitaev's honeycomb spin liquid [3], [4], [5]. Its inelastic neutron scattering (INS) spectrum shows a broad continuum whose origin is debated [6], [7], [8], [9]. Several models of its magnetic interactions compete [9], [10], [11], and exact calculations on small clusters are the usual way to compare them with data [11]. This makes it a natural target for quantum simulation, and a natural test of when such a simulation would pay off.

Contributions. (i) An accuracy target taken from the energy resolution of INS. The time window, the total simulated time and the allowed error all follow from this one number (Sec. II). (ii) The measured classical cost of computing $S(\mathbf{q},\omega)$ exactly on periodic clusters, including clusters with several ground states, and of MPS simulation, whose bond dimension we check against exact results (Sec. III). (iii) Trotter errors measured on the spectrum itself, for the same circuit whose gates we count, and a full surface-code resource estimate, with ground-state preparation counted separately (Sec. IV). (iv) The crossover size N^*, how much it depends on our assumptions, and how far the hardware it needs is from India's stated targets (Sec. V).

II. Model and accuracy target

We use the spin-1/2 J_1-K-Γ-Γ'-J_3 model on the honeycomb lattice [12], [9],

$$H = \sum_{\langle ij\rangle_\gamma} \Big[J_1\, \mathbf{S}_i\cdot\mathbf{S}_j + K S_i^\gamma S_j^\gamma + \Gamma\big(S_i^\alpha S_j^\beta + S_i^\beta S_j^\alpha\big) + \Gamma' \sum_{\mu=\alpha,\beta} \big(S_i^\gamma S_j^\mu + S_i^\mu S_j^\gamma\big) \Big] + J_3 \sum_{\langle\langle\langle ij\rangle\rangle\rangle} \mathbf{S}_i\cdot\mathbf{S}_j, \quad (1)$$

where $\gamma \in \{x,y,z\}$ labels the bond and α,β are the other two axes. Our main parameter set is the minimal model of Winter *et al.* [9], fitted to ab initio results and neutron data: $(J_1, K, \Gamma, J_3) = (-0.5, -5.0, 2.5, 0.5)$ meV (as listed in [13]).

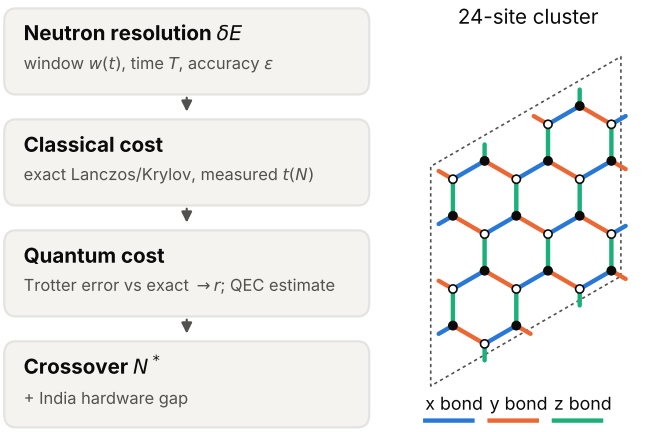


Fig. 1. The method. The neutron resolution sets the time window, the total time and the accuracy. We find the classical and quantum costs for the same target, and they meet at N^*. Right: a 24-site periodic cluster with Kitaev bond labels. Open and filled circles are the two sublattices; short bond stubs leave the cell and come back on the opposite side.

Two more sets test how much the model matters: the K-Γ model of Ran *et al.*, $(K, \Gamma) = (-6.8, 9.5)$ meV [10], and the set that Laurell and Okamoto fitted to the specific heat [11], $(J_1, K, \Gamma, \Gamma', J_3) = (-1.3, -15.1, 10.1, -0.12, 0.9)$ meV (as listed in [14]). Their energy scales are about two and three times larger.

Observable. We compute

$$S(\mathbf{q},\omega) = \frac{1}{2\pi}\int_{-T}^{T} dt\, w(t)\, e^{i\omega t}\, \frac{1}{N}\sum_{ij} e^{i\mathbf{q}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \mathbf{S}_i(t)\cdot\mathbf{S}_j(0)\rangle \tag{2}$$

with $\hbar = 1$ and energies in meV. An instrument with a Gaussian energy resolution of full width δE measures the true spectrum blurred by that Gaussian. In time, this blurring is exactly the window $w(t) = e^{-t^2/2\tau^2}$ with $\tau = 2\sqrt{2\ln 2}/\delta E$. So we only need the correlations up to the time $T = \tau\sqrt{2\ln(1/c)}$, where the window drops below $c = 10^{-3}$. Sampling every $\Delta t = \pi/\omega_{\max}$ covers energies up to $\omega_{\max} = 30$ meV. We take $\delta E = 0.50$ meV as a typical high-resolution value; Table II shows coarser ones. This gives $T = 17.5\,\text{meV}^{-1} = 11.5$ ps and 169 time points (the rough estimate $2\pi\hbar/\delta E$ gives 8.3 ps). We require the blurred spectrum $\sum_a S^{aa}(\mathbf{q},\omega)$ to be correct to $\varepsilon = 5\%$ of its peak, a typical experimental uncertainty (Table II covers 1–10%). Half of this error budget goes to Trotter error and half to shot noise.

III. Classical cost

Clusters. For each N we list all periodic clusters of $N/2$ unit cells and pick the most compact one in which every first- and third-neighbour bond joins a different pair of sites. Third-neighbour bonds only work this way for $N \geq 14$. We use $N = 14, 16, 18$ for the Winter set and $N = 14, 16$ for the other two; larger clusters are left for future work.

Method. The model conserves no spin component, so we work with the full space of 2^N states and apply the Hamiltonian without storing it. We find the ground state with the Lanczos method. Symmetric clusters have several ground states: 6 for $N = 14$ (three zigzag directions times time reversal) and 2 for $N = 18$ with the Winter parameters. We find all of them and average over them, because a single one breaks the translation symmetry. We evolve each $S^b_j|0\rangle$ in time with the short-iterative Lanczos method [15], with an error of at most 10^{-9} per step, and read off all correlations at each time. Translation symmetry means one reference site j per sublattice is enough. We tested the code on the exactly solvable Kitaev model [3], [16], where it reproduces the conserved plaquettes, the correlations that vanish beyond nearest neighbours, and the ground-state energy to within 2%. For $N = 8$, the spectrum agrees with an exact eigenstate calculation to 10^{-4} of its peak.

Measured cost. The full spectrum takes 7.1 min, 5.1 min and 49 min on a 4-core workstation for $N = 14$, 16 and 18, counting all ground states. Per ground state the cost grows as $N^2 2^N$: each product with the Hamiltonian costs about $N2^N$, and the number of products grows as N. A free exponential fit gives a growth factor of 2.13 per spin. Fig. 2 shows the spectrum for $N = 18$.

Matrix product states. We also evolve $S^z_j|0\rangle$ with two-site TDVP [17] in TeNPy [18], starting from a DMRG ground state, on cylinders with a circumference of $W = 2$ and 3 unit cells (16 and 18 spins). The MPS code uses the same bonds as the exact code, and at full bond dimension the two agree to 10^{-9}. For each bond dimension χ we find the longest time window T' for which the error of the blurred spectrum stays below ε. We bound this error by summing the errors of all site pairs. For $W = 2$, $\chi = 16$, 32, 64 and 128 reach $T' = 1.2$, 3.1, 12.4 and 17.5 meV^{-1} (the full window). For $W = 3$, $\chi = 32$ and 64 reach only 1.1 and 2.5 meV^{-1}. A fit of $\ln\chi = c_W + s_W T'$ gives growth rates $s_W = 0.11$ and 0.51 meV, so the full window at $W = 3$ would already need $\chi \approx 1.4 \times 10^5$. The cost per step grows as $ND\chi^3$, where D is the bond dimension of the Hamiltonian operator. We extrapolate it to cylinders with $N = 2W^2$, taking s_W and c_W linear in W and six runs (three spin components, two sublattices) per spectrum.

IV. Quantum cost

Circuit. We use one qubit per spin and one extra (ancilla) qubit. Each correlation is measured with the Hadamard test. The ancilla controls only two Pauli gates, and the time evolution between them runs without control [19]. So the time evolution is the only part that needs costly non-Clifford gates. It is a second- or fourth-order Suzuki product formula [20] built from the Pauli terms of H. Each term is a rotation $e^{-i\theta\sigma^a_i\sigma^b_j}$. There are five such rotations per nearest-neighbour bond for the Winter set (nine if $\Gamma' \neq 0$) and three per third-neighbour bond. We group them by bond type and merge neighbouring rotations about the same term.

Trotter error, measured. We simulate exactly this sequence of rotations on a state vector and form the correlations as the Hadamard test would, evolving both $|0\rangle$ and $S^b_j|0\rangle$ with

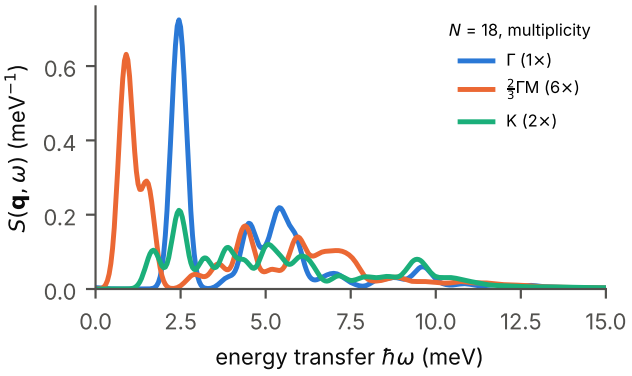


Fig. 2. The computed spectrum $\sum_a S^{aa}(\mathbf{q},\omega)$ at the distinct momenta of the 18-site cluster (Winter parameters), blurred by the resolution $\delta E = 0.50\,\text{meV}$. The legend gives each momentum and how many equivalent momenta it stands for.

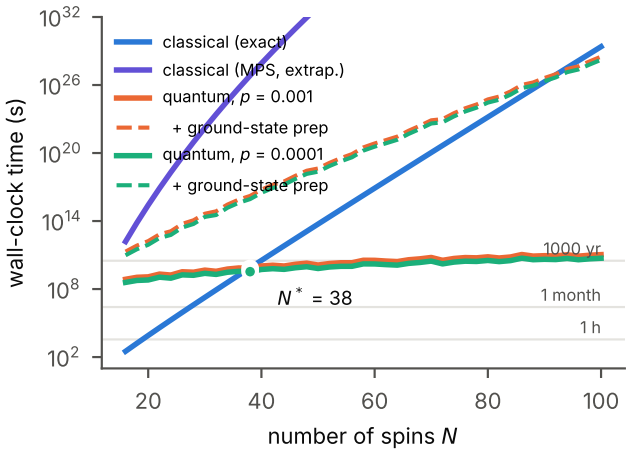


Fig. 3. Time to compute $S(\mathbf{q},\omega)$ at the experimental resolution and accuracy, against the number of spins. Classical: exact simulation, measured for $N \le 18$ and extrapolated as $N^2 2^N$, and MPS simulation extrapolated from cylinders with $W = 2$ and 3 (it leaves the plot near $N = 30$). Quantum: second-order Trotter circuits on a surface code, without (solid) and with (dashed) ground-state preparation.

the circuit. We then compare the blurred spectrum with the exact one for $N = 14, 16, 18$, for several step sizes δt and three resolutions. The error follows $\varepsilon_{\text{Trot}} = A\,T\,\delta t^p$, with $p = 2.00$ for second order (30 points). Fitting an extra factor N^α gives $\alpha = -0.12$, so the error does not grow with N. We use the largest prefactor among the sizes we measured, $A = 0.73$. This is what we expect for a quantity that does not grow with system size and is limited by the resolution. Worst-case commutator bounds [21] would give far larger numbers.

Shots. Each shot gives ± 1, so after n shots the variance of a correlation of spin-1/2 operators is at most $1/16n$. Carried through the windowed transform, this gives $\text{Var}\,S \le 3N\sum_k (w_k \Delta t)^2/(32\pi^2 n)$. We choose n so that two standard deviations stay below half of $\varepsilon S_{\max}$. Here $S_{\max}$ is the peak of the inelastic spectrum ($\omega > 2\delta E$). It leaves out the weight near zero energy that builds up at the zigzag ordering wavevector on clusters that fit this order, and it takes the smallest peak over the sizes we measured. For the Winter set, $S_{\max} = 0.23\,\text{meV}^{-1}$ and $n \approx 1.9 \times 10^4$. With one reference site per sublattice, three spin components, and real and imaginary parts, we need $12N$ circuits per time point, or 7.7×10^4 in total at $N = 38$. All are independent and could run on many machines at once. We compare one quantum computer with one classical workstation.

Error correction. We pass the logical counts (qubits, rotations, and rotation depth from scheduling each rotation as early as possible) to Microsoft's resource estimator [22]. We use a surface code [23], gate-based qubits with nanosecond gates, physical error rates $p = 10^{-3}$ and 10^{-4}, and a total error budget of 1% per circuit. The estimator includes rotation synthesis [24] and magic-state factories. One machine runs all circuits one after another.

Ground-state preparation. We use a simple model: phase estimation starting from a simple product state with overlap γ. Each attempt takes an evolution time $2\pi \ln(1/\delta)/\Delta$, controlled evolution costs twice as much, and each shot needs $1/\gamma^2$ attempts. Better methods need only $1/\gamma$ [25]. From exact results we take the mean gap $\Delta = 0.37\,\text{meV}$ and the best product-state overlap, optimised over all spin directions. The overlap falls as $\gamma \propto e^{-0.21N}$ over $N = 14, 16, 18$. We report this cost separately because it is the least certain part.

TABLE I
FAULT-TOLERANT RESOURCES AT THE CROSSOVER FOR THE THREE PARAMETER SETS (SECOND-ORDER TROTTER, $\delta E = 0.50\,\text{MEV}$, $\varepsilon = 5\%$). r: TROTTER STEPS FOR THE LONGEST TIME. d: CODE DISTANCE. THE RUNTIME LEAVES OUT GROUND-STATE PREPARATION.

Set	p	N^*	r (steps)	T states per shot	d	Phys. qubits	Runtime at N^*
Winter	10^{-3}	38	396	7.2×10^6	19	1.7 M	209 yr
Winter	10^{-4}	38	396	7.2×10^6	9	0.25 M	109 yr
Ran	10^{-3}	36	1757	1.1×10^7	19	5.2 M	55 yr
Ran	10^{-4}	36	1757	1.1×10^7	9	0.41 M	28 yr
Laurell	10^{-3}	38	2751	7.9×10^7	21	3.5 M	558 yr
Laurell	10^{-4}	38	2751	7.9×10^7	11	0.60 M	270 yr

V. Crossover and India roadmap

Fig. 3 shows the result for the Winter set. The quantum cost grows only as a power of N, but it starts high. At N^* one circuit runs for seconds, and about 10^5 circuits each need about 10^4 shots. The curves cross at $N^* = 38$ for both error rates. There the quantum computer needs 109 yr ($p = 10^{-4}$) or 209 yr ($p = 10^{-3}$), against 241 yr for the classical computer (Table I). So the crossover tells us about scaling, not about a useful calculation. A practical simulation at N^* needs much more parallel hardware and better algorithms on both sides. Fourth-order formulas give the same N^*, because they need fewer steps but five times as many rotations per step. With ground-state preparation the crossover moves to $N^* = 92$–94, because the product-state overlap shrinks exponentially. Better starting states [25] would help most. The extrapolated MPS

TABLE II

HOW THE CROSSOVER DEPENDS ON OUR ASSUMPTIONS. EACH ROW CHANGES ONE ASSUMPTION FROM THE BASELINE (WINTER SET, $\delta E = 0.50$ MEV, $\varepsilon = 5\%$, SECOND-ORDER TROTTER, NANOSECOND GATES WITH $p = 10^{-4}$, ONE CLASSICAL WORKSTATION). RUNTIME AND PHYSICAL QUBITS ARE FOR THE QUANTUM COMPUTER AT N^*, WITHOUT GROUND-STATE PREPARATION.

Variation	N^*	Runtime	Phys. qubits
Baseline (§V)	38	109 yr	0.25 M
Accuracy $\varepsilon = 1\%$	44	9927 yr	0.24 M
Accuracy $\varepsilon = 2\%$	42	2005 yr	0.19 M
Accuracy $\varepsilon = 10\%$	36	20 yr	0.2 M
Resolution $\delta E = 1$ meV	36	25 yr	0.2 M
Resolution $\delta E = 2$ meV	34	2 yr	0.12 M
Physical error $p = 10^{-3}$	38	209 yr	1.7 M
μs gates, $p = 10^{-4}$	50	308374 yr	0.31 M
μs gates, $p = 10^{-3}$	50	628735 yr	1.1 M
Fourth-order Trotter	38	89 yr	0.25 M
With ground-state prep.	92	n/a	n/a
Classical $10^2\times$ faster	44	173 yr	0.24 M
Classical $10^4\times$ faster	52	278 yr	0.26 M
Ran parameters	36	28 yr	0.41 M
Laurell and Okamoto parameters	38	270 yr	0.6 M

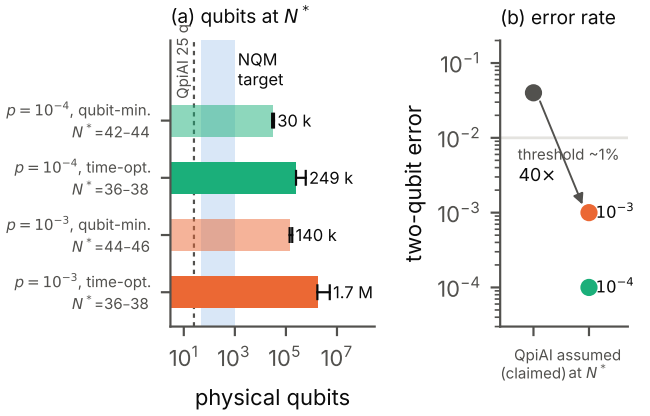


Fig. 4. (a) Physical qubits needed at the crossover, for the fastest and for the smallest surface-code layout, at two physical error rates. The shaded band is the National Quantum Mission target of 50 to 1000 physical qubits [26], and the dashed line is the 25-qubit QpiAI-Indus system [27]. Bars show the Winter set and whiskers the range over the three parameter sets. (b) The assumed physical error rates against the reported two-qubit error of QpiAI-Indus.

cost is higher than the exact cost at every size we consider (Fig. 3). The quantum computer beats it from the smallest size on, but it is never the cheaper classical method. The long time window that the neutron resolution demands makes entanglement grow, and this, not the system size, limits MPS. So exact simulation sets the crossover.

Sensitivity. Table II changes one assumption at a time. Over all 192 combinations (three parameter sets, three resolutions, four accuracies, four qubit types, and both Trotter orders where measured), N^* stays between 32 and 56 against one workstation. *Experiment:* a tighter accuracy or a finer resolution moves N^* by only a few spins. But it changes the quantum run time at N^* hugely, from years at $\delta E = 2$ meV to millennia at $\varepsilon = 1\%$, because the number of shots grows as ε^{-2} and the circuits get longer with T. *Quantum hardware:* the physical error rate sets the number of qubits but barely moves N^*. Slow gates (microseconds), as in trapped ions, make each shot about 1500 times longer and give the largest N^* values. *Classical hardware:* each doubling of classical speed adds roughly one spin, so a large cluster 10^4 times faster than our workstation raises N^* by about 14. *Model:* the Ran *et al.* and Laurell and Okamoto sets (exact data for $N = 14, 16$) have Trotter prefactors $A = 16$ and 24, twenty to thirty times that of the Winter set. Yet they cross over at nearly the same N^*, because the number of steps grows only as $A^{1/2}$, and the circuit and shot counts, which dominate, barely change. With only two sizes we cannot tell how their overlap changes with N (for the Laurell and Okamoto set it even rises from $N = 14$ to 16). So we give the crossover with ground-state preparation for the Winter set only.

India roadmap. The National Quantum Mission aims for machines with 50 to 1000 physical qubits within eight years [26]. The superconducting QpiAI-Indus system has 25 qubits with a reported two-qubit fidelity of 96% [27]. We work out the hardware needed at the crossover for all three parameter sets and two surface-code layouts (Fig. 4). The fastest layout, used in Table I, adds as many magic-state factories as speed up the run. The smallest layout uses one factory. Each shot then takes 60 to 75 times longer, so the crossover comes later. *Qubits.* At $p = 10^{-4}$ the crossover needs 249–597 thousand physical qubits in the fastest layout ($N^* = 36$–38) and 29–34 thousand in the smallest one ($N^* = 42$–44, but 5512–38363 years of run time). At $p = 10^{-3}$ these numbers become 1708–5218 and 140–177 thousand. The smallest of these machines is 29–34 times larger than the mission's 1000-qubit target. That is about 5 doublings of qubit count beyond 2031, and the fastest layout is 8–9 doublings away. *Error rate.* A fidelity of 96% means a 4% error. This is 4 times above the surface-code threshold of about 1%, so no number of such qubits can run this calculation. Reaching $p = 10^{-3}$ or 10^{-4} needs an error 40 or 400 times lower. The error rate is therefore the main limit and comes first. Below threshold, each tenfold drop in p cuts the qubit count by a factor of five to thirteen in our estimates, which is worth more than any of the doublings above. For India's roadmap, this application asks first for two-qubit errors near 10^{-4}. Only then does it ask for 10^4 to 10^5 qubits, well beyond the 2031 target.

Hardware check. To see where today's hardware stands, we run the same Trotter circuit on 12 qubits. The full Hadamard-test method, with ground-state preparation, needs far too many gates. So we use a standard test instead: we start an open 12-spin honeycomb flake with the Winter couplings in the Néel state, evolve it, and measure every spin along z. The Qiskit [28] circuit gives exactly the same result as our state-vector Trotter code. Compiled for IBM's Heron processor `ibm_torino`, it needs 180 two-qubit gates per Trotter step. On an emulator with that processor's published noise cali-

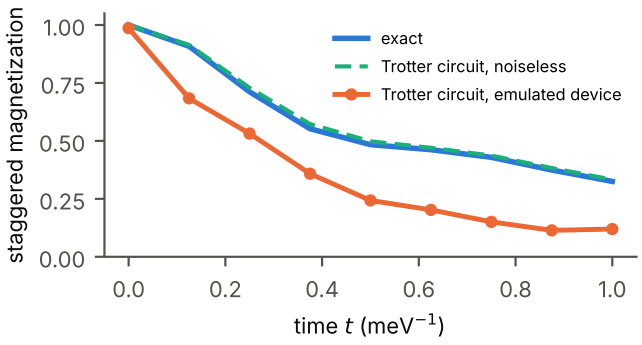


Fig. 5. Hardware check on 12 qubits. The staggered magnetization after starting a 12-spin flake (Winter couplings) in the Néel state, from exact evolution, from the noiseless Trotter circuit, and from the same circuit compiled for `ibm_torino` and run on an emulator with that processor's published noise calibration (2000 shots per point, one Trotter step per $0.125\,\mathrm{meV}^{-1}$, and 180 to 1353 two-qubit gates).

bration (not yet a device run), the measured magnetizations differ from the exact ones by up to 0.39 after a single step at $t = 0.125\,\mathrm{meV}^{-1}$, only 0.7% of the window T. The noiseless circuit stays within 0.04. After 1353 two-qubit gates, only 37% of the staggered magnetization survives (Fig. 5). Without error correction, the accuracy the experiment needs is out of reach by orders of magnitude in circuit depth. This is why the crossover assumes a fault-tolerant machine.

VI. Limitations and Outlook

Classical baseline. Exact simulation is not the strongest possible classical method, so our N^* is a lower limit on where an advantage could start. Translation symmetry would save another factor of about $N/2$, and continued-fraction Lanczos [29] avoids time stepping altogether for a fixed blurring. Our MPS estimate is rough. It rests on two short cylinders, an upper bound on the error rather than the error in momentum space, and a straight-line fit of $\ln \chi$ in time and width. We did not run PEPS, time-dependent variational Monte Carlo or neural quantum states. Each could raise N^*, although long-time dynamics in two dimensions is where they struggle most. *Extrapolation.* We extrapolate the classical costs and Trotter errors from $N \le 18$ for the Winter set and $N \le 16$ for the others. Our claim that the Trotter error does not grow with N rests on few sizes with strong finite-size effects, and so does the reference peak $S_{\max}$, which varies threefold between sizes. *Measurement.* Measuring $S(\mathbf{q}, t)$ directly instead of correlations between site pairs, or sharing shots between circuits, could cut the number of circuits by up to a factor of N. *Physics.* We assume zero temperature, ignore the neutron polarization factor, and use parameter sets that are themselves uncertain. *Ground-state preparation* is modelled roughly and dominates when included. Even so, tying the accuracy to an experiment turns "quantum advantage for materials" into a concrete number that others can check, and our code lets each assumption be changed.

Code and data will be released publicly upon acceptance.

Acknowledgment

Code, figures and draft text for this work were prepared with the help of an AI system (Claude, Anthropic). The authors checked the methods, results and text and take responsibility for them.

References

[1] N. Yoshioka, T. Okubo, Y. Suzuki, Y. Koizumi, and W. Mizukami, "Hunting for quantum-classical crossover in condensed matter problems," *npj Quantum Inf.*, vol. 10, 2024, doi:10.1038/s41534-024-00839-4.

[2] J. Sun *et al.*, "Quantum-classical crossover in fault-tolerant quantum dynamics simulation," 2026, arXiv:2607.16116.

[3] A. Kitaev, "Anyons in an exactly solved model and beyond," *Ann. Phys.*, vol. 321, pp. 2–111, 2006.

[4] G. Jackeli and G. Khaliullin, "Mott insulators in the strong spin-orbit coupling limit: From Heisenberg to a quantum compass and Kitaev models," *Phys. Rev. Lett.*, vol. 102, p. 017205, 2009.

[5] H. Takagi, T. Takayama, G. Jackeli, G. Khaliullin, and S. E. Nagler, "Concept and realization of Kitaev quantum spin liquids," *Nat. Rev. Phys.*, vol. 1, pp. 264–280, 2019.

[6] A. Banerjee *et al.*, "Proximate Kitaev quantum spin liquid behaviour in a honeycomb magnet," *Nat. Mater.*, vol. 15, pp. 733–740, 2016.

[7] A. Banerjee, J. Yan, J. Knolle, C. A. Bridges, M. B. Stone, M. D. Lumsden, D. G. Mandrus, D. A. Tennant, R. Moessner, and S. E. Nagler, "Neutron scattering in the proximate quantum spin liquid α-$RuCl_3$," *Science*, vol. 356, pp. 1055–1059, 2017.

[8] S.-H. Do, S.-Y. Park, J. Yoshitake, J. Nasu, Y. Motome, Y. S. Kwon, D. T. Adroja, D. J. Voneshen, K. Kim, T.-H. Jang, J.-H. Park, K.-Y. Choi, and S. Ji, "Majorana fermions in the Kitaev quantum spin system α-$RuCl_3$," *Nat. Phys.*, vol. 13, pp. 1079–1084, 2017.

[9] S. M. Winter, K. Riedl, P. A. Maksimov, A. L. Chernyshev, A. Honecker, and R. Valentí, "Breakdown of magnons in a strongly spin-orbital coupled magnet," *Nat. Commun.*, vol. 8, p. 1152, 2017.

[10] K. Ran, J. Wang, W. Wang, Z.-Y. Dong, X. Ren, S. Bao, S. Li, Z. Ma, Y. Gan, Y. Zhang, J. T. Park, G. Deng, S. Danilkin, S.-L. Yu, J.-X. Li, and J. Wen, "Spin-wave excitations evidencing the Kitaev interaction in single crystalline α-$RuCl_3$," *Phys. Rev. Lett.*, vol. 118, p. 107203, 2017.

[11] P. Laurell and S. Okamoto, "Dynamical and thermal magnetic properties of the Kitaev spin liquid candidate α-$RuCl_3$," *npj Quantum Mater.*, vol. 5, p. 2, 2020.

[12] J. G. Rau, E. K.-H. Lee, and H.-Y. Kee, "Generic spin model for the honeycomb iridates beyond the Kitaev limit," *Phys. Rev. Lett.*, vol. 112, p. 077204, 2014.

[13] P. A. Maksimov and A. L. Chernyshev, "Rethinking α-$RuCl_3$," *Phys. Rev. Research*, vol. 2, p. 033011, 2020.

[14] "Semiclassical routes to the α-$RuCl_3$ scattering continuum via model meta-analysis," 2026, arXiv:2602.10190.

[15] T. J. Park and J. C. Light, "Unitary quantum time evolution by iterative Lanczos reduction," *J. Chem. Phys.*, vol. 85, pp. 5870–5876, 1986.

[16] G. Baskaran, S. Mandal, and R. Shankar, "Exact results for spin dynamics and fractionalization in the Kitaev model," *Phys. Rev. Lett.*, vol. 98, p. 247201, 2007.

[17] J. Haegeman, C. Lubich, I. Oseledets, B. Vandereycken, and F. Verstraete, "Unifying time evolution and optimization with matrix product states," *Phys. Rev. B*, vol. 94, p. 165116, 2016.

[18] J. Hauschild and F. Pollmann, "Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy)," *SciPost Phys. Lect. Notes*, p. 5, 2018.

[19] R. Somma, G. Ortiz, J. E. Gubernatis, E. Knill, and R. Laflamme, "Simulating physical phenomena by quantum networks," *Phys. Rev. A*, vol. 65, p. 042323, 2002.

[20] M. Suzuki, "General theory of fractal path integrals with applications to many-body theories and statistical physics," *J. Math. Phys.*, vol. 32, pp. 400–407, 1991.

[21] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, "Theory of Trotter error with commutator scaling," *Phys. Rev. X*, vol. 11, p. 011020, 2021.

[22] M. E. Beverland, P. Murali, M. Troyer, K. M. Svore, T. Hoefler, V. Kliuchnikov, G. H. Low, M. Soeken, A. Sundaram, and A. Vaschillo, "Assessing requirements to scale to practical quantum advantage," 2022, arXiv:2211.07629.

[23] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, "Surface codes: Towards practical large-scale quantum computation," *Phys. Rev. A*, vol. 86, p. 032324, 2012.
[24] N. J. Ross and P. Selinger, "Optimal ancilla-free Clifford+T approximation of z-rotations," *Quantum Inf. Comput.*, vol. 16, pp. 901–953, 2016.
[25] L. Lin and Y. Tong, "Near-optimal ground state preparation," *Quantum*, vol. 4, p. 372, 2020.
[26] Department of Science and Technology, Government of India, "National quantum mission," https://dst.gov.in/national-quantum-mission-nqm, 2023, approved 19 April 2023.
[27] QpiAI, "QpiAI announces dawn of quantum era in India with 25 qubit quantum computer," Business Wire press release, 15 April 2025, 2025.
[28] A. Javadi-Abhari *et al.*, "Quantum computing with Qiskit," 2024, arXiv:2405.08810.
[29] E. R. Gagliano and C. A. Balseiro, "Dynamical properties of quantum many-body systems at zero temperature," *Phys. Rev. Lett.*, vol. 59, pp. 2999–3002, 1987.